

Final Year Project Proposal

DEPARTMENT OF COMPUTER SCIENCE

1. Project Title

AI Virtual Doctor: A Multi-Modal Integrated Diagnostic Engine for 34 Clinical Conditions Using Un-Nested Fine-Tuned MobileNetV2 and Late-Fusion Explainable AI (XAI)

2. Student Information

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4. Project Domain / Area

- Artificial Intelligence
- Machine Learning / Deep Learning
- Data Science / Big Data
- Computer Vision / Image Processing

5. Problem Statement

The healthcare sector in Pakistan is currently grappling with a severe '**Triage Gap**' that disproportionately affects rural and underserved urban populations. This crisis is primarily driven by significant reductions in federal and developmental health budgets, leading to a state of chronic resource overload in public hospitals. The resulting environment is characterized by a critical shortage of specialized medical personnel, leaving general practitioners and frontline workers overwhelmed by the sheer volume of patients requiring urgent diagnosis.

This institutional strain has created a dangerous reliance on **unimodal diagnostic tools**—processes that often consider only patient symptoms while neglecting critical visual or chemical evidence. Clinically, many high-prevalence diseases in Pakistan, such as **Tuberculosis (TB), Pneumonia, and Malaria**, present with overlapping clinical features like fever and cough. Without the integration of 'Visual Evidence' from imaging (X-rays, Ultrasounds) and 'Chemical Evidence' from lab markers, these conditions cannot be accurately distinguished, leading to high rates of 'mis-triage' and incorrect treatment paths.

Furthermore, current Artificial Intelligence (AI) implementations in the medical field frequently operate as '**Black Boxes**'. These models provide a diagnosis without explaining the underlying reasoning, which creates a significant trust deficit among both clinicians and patients. This lack of transparency, combined with a lack of accessible specialized care, has contributed to a rise in dangerous self-medication practices among the public.

A specific technical challenge identified in previous iterations of this project was the "**Visual Context Unavailable**" error. This occurred because the AI model's "eyes" (the image processing branch) were effectively turned off or "frozen," causing the system to ignore crucial medical imagery and default to a diagnosis based solely on symptoms—such as the persistent "Typhoid" prediction even when presented with a clear Radiology scan.

To address these systemic and technical failures, there is an urgent need for a **multimodal, explainable diagnostic engine**. This engine must force the integration of deep visual analysis with clinical data to bridge the triage gap, restore clinical transparency through heatmaps, and provide a reliable decision-support tool for resource-constrained environments.

6. Objectives

The primary goal of this project is to architect a next-generation diagnostic support system that addresses the technical and clinical shortcomings of traditional unimodal AI. The expanded objectives are as follows:

- **Engineering an Un-Nested Multimodal Late-Fusion Pipeline:** To move beyond simple sequential models by developing a Functional API-based architecture that processes image and tabular symptom data through independent wings. This ensures that the model can weigh "Visual Evidence" (imaging) and "Clinical Evidence" (symptoms) simultaneously before reaching a unified diagnostic conclusion for 34 distinct conditions.
- **Deep Feature Extraction via Fine-Tuned MobileNetV2:** To transition from frozen transfer learning to a deep fine-tuning strategy. By unfreezing the top layers of the MobileNetV2 base and setting a specialized low learning rate ($1e-5$), the objective is to force the model to learn intricate medical textures—such as pulmonary infiltrates or dermatological patterns—rather than relying on pre-trained general patterns.
- **Restoration of Clinical Transparency through Explainable AI (XAI):** To resolve the "Hidden Layer" visibility issues found in previous nested models. By utilizing an un-nested architecture, the system will provide direct access to the convolutional feature maps, enabling the generation of Grad-CAM heatmaps that highlight the specific anatomical "Zone of Infection" for physician verification.
- **Large-Scale Diagnostic Scaling (34 Conditions):** To expand the diagnostic scope from a limited 14-class system to a comprehensive 34-class engine. This involves creating a specialized stratified data pipeline to handle 156,000+ medical files across Radiology, Pathology, Sonography, and Dermatology wings.
- **Development of an Automated OCR Clinical Scanner:** To integrate Optical Character Recognition (OCR) technology using the Tesseract engine to bridge the gap between physical medical records and digital AI. The objective is to allow the system to "read" printed reports, extract key clinical symptoms automatically, and reduce the risk of human transcription errors during the triage process.
- **Optimization for Resource-Constrained Environments:** To ensure high-performance diagnostic accuracy remains achievable on local hardware (standard laptops) without requiring expensive cloud-based GPU clusters. This is achieved by implementing efficient architectures and custom data generators that optimize memory usage for the oneDNN-enabled environment.

7. Scope of the Project

Included: Preliminary diagnosis for **34 diseases** across four specialized wings:

- **Radiology:** COVID-19, TB, Pneumonia, Asthma, Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural, Pneumothorax.
- **Pathology/Sonography/Dermatology:** Malaria, Dengue, Typhoid, Hepatitis, Jaundice, Cirrhosis, Bacterial Infections, Fungal Infections, Parasitic Infections, Liver Disease, and Gallbladder Diseases.

8. Proposed Solution / Methodology

The methodology is divided into three primary pipelines that ensure the model balances clinical data with deep image analysis.

1. Multimodal Data Pipeline & Standardization

The system processes two distinct data streams simultaneously:

- **Visual Stream (Image Branch):** Medical images (X-rays, Ultrasounds, Microscopy) are standardized to a 224 $\times$ 224 pixel resolution. We implement **Real-Time Data Augmentation**—including horizontal flips, brightness adjustments, and contrast shifts—to simulate the variability of images captured via mobile phone cameras for the future "Scanner" module.
- **Clinical Stream (Symptom Branch):** Tabular data covering six key symptoms (Fever, Cough, Nausea, Abdominal Pain, Shortness of Breath, Rash) is vectorized into a binary or normalized numeric format.

2. Un-Nested Functional API Architecture

Unlike standard "Sequential" models, this project utilizes the **Keras Functional API** to create a "Late Fusion" architecture.

- **Base Model:** We utilize **MobileNetV2** (previously EfficientNetB0) due to its superior performance-to-size ratio on local hardware.
- **The "Un-Nesting" Fix:** We explicitly define the input and output tensors of the MobileNet base (base.input to base.output). This ensures that the **Grad-CAM engine** can access internal convolutional layers (like out_relu) without being blocked by a "hidden" container.
- **Fusion Layer:** The feature vectors from both the Image Wing and Clinical Wing are concatenated into a single "Master Vector" before passing through final dense layers for a 34-class Softmax classification.

3. Fine-Tuning & Weight Balancing Strategy

To ensure the model doesn't ignore images (the "Typhoid Bias"), we utilize a two-stage training strategy:

- **Unfreezing the Base:** We set `base.trainable = True`, specifically allowing the top 30+ layers of the MobileNet base to adapt to medical-specific textures.
- **Optimized Learning Rate:** We employ an extremely low learning rate ($1e-5$) using the Adam optimizer to prevent the "Catastrophic Forgetting" of pre-trained ImageNet features while slowly learning 34 new medical classes.

9. Expected Outcomes

The successful completion of this project is expected to yield a robust, transparent, and clinically viable diagnostic system. The outcomes are categorized into technical artifacts, clinical impact, and documentation:

1. High-Fidelity Multimodal Diagnostic Model

- **34-Class Specialized Model:** A fully trained .h5 model capable of identifying 34 distinct medical conditions with a balanced reliance on both visual and clinical data.
- **Corrected Visual Weighting:** A system that has successfully overcome the "Typhoid Bias" by utilizing fine-tuned, un-nested convolutional layers to prioritize medical imagery when provided.

2. Transparent and Explainable Interface

- **XAI-Enabled Heatmaps:** A user interface that generates Grad-CAM heatmaps, providing a visual justification for each diagnosis by highlighting the specific "Zone of Infection" within the scan.
- **Differential Diagnosis Output:** Rather than a single "Black Box" result, the system will provide the top three most likely conditions (Differential Diagnosis) to assist clinicians in considering alternative medical possibilities.

3. Automated Data Entry via OCR

- **Clinical Report Scanner:** A functional Optical Character Recognition (OCR) module that can extract key symptoms (Fever, Cough, etc.) from physical, printed medical reports and automatically populate the model's input fields.

- **Reduced Manual Error:** By automating the entry of clinical markers, the system minimizes the risk of human error during the triage process.

4. Clinical and Social Impact

- **Reduction in 'Mis-Triage' Rates:** By integrating image analysis with symptoms, the system aims to significantly lower the rate at which overlapping conditions (like TB and Pneumonia) are confused in resource-strained settings.
- **Bridging the Triage Gap:** A portable decision-support tool that empowers frontline healthcare workers in rural Pakistan to make faster, evidence-based referral decisions.

5. Comprehensive Technical Documentation

- **Research Thesis:** A detailed project report documenting the transition from Sequential to Functional API architectures and the impact of unfreezing MobileNetV2 on diagnostic accuracy.
- **Performance Metrics:** Objective validation of the model using an 80/20 data split, providing clear metrics on precision, recall, and F1-score across all 34 medical categories.

10. Timeline (Milestones)

The timeline has been adjusted to account for the increased complexity of managing 156,000+ images and the technical requirements of un-nesting the functional API for Explainable AI (XAI).

Phase	Task	Duration	Expected Completion
Phase 1	Requirement Analysis & Big Data Acquisition: Finalizing the expanded dataset of 156,000+ images across Radiology, Pathology, Sonography, and Dermatology wings.	2 weeks	Week 2
Phase 2	Advanced Preprocessing & Augmentation: Standardizing images to \$224 \times 224\$ and	3 weeks	Week 5

Phase	Task	Duration	Expected Completion
	implementing 34-class stratified splitting (80/20) to ensure balanced training.		
Phase 3	Un-Nested Architecture & Fine-Tuning: Implementing the Functional API, unfreezing the MobileNetV2 base (trainable=True), and integrating Grad-CAM heatmaps for 34 conditions.	6 weeks	Week 11
Phase 4	Hybrid Portal Development & OCR Integration: Building the diagnostic interface and the Tesseract-based clinical report scanner for automated symptom entry.	3 weeks	Week 14
Phase 5	Documentation, Validation & Final Submission: Finalizing the 4-page thesis, performing system stress tests on local hardware, and final defense preparation.	2 weeks	Week 16

Key Technical Adjustments in this Timeline:

- **Phase 2 Expansion:** Increased time allocated for **Stratified Splitting** to ensure that smaller classes among the 34 conditions are not ignored by the model.
- **Phase 3 Depth:** The 6-week window now specifically includes the "**Un-Nesting**" process, which is critical for restoring heatmap visibility that was previously blocked by hidden layers.
- **Phase 4 Innovation:** Added the **OCR (Scanner) module** development, which allows the project to move beyond manual symptom entry to automated report processing.

11. Resources Required

- **Hardware:** Local workstation (HP) with optimized oneDNN CPU/GPU support for processing the expanded 156,000+ image dataset.
- **Software:** Python 3.12, TensorFlow 2.10+, and Tesseract OCR for automated clinical report scanning.

12. Risk Factors & Limitations

1. Computational Constraints & Local Hardware Risks

- **Risk:** Processing 156,000+ images on local laptop hardware (HP/Toshiba) creates a risk of **Thermal Throttling** and **Out-of-Memory (OOM)** errors.
- **Mitigation:** We utilize **MobileNetV2** for its low parameter count and a custom **Hybrid Data Generator** that loads data in small batches (32 images at a time) rather than loading the entire dataset into RAM.

2. Data Bias & "Black Box" Trust Deficit

- **Risk:** If the dataset contains more samples for one disease (e.g., Typhoid), the model may develop a bias, leading to "Mis-triage".
- **Mitigation:** We implement **Stratified Splitting** (80/20) and **Explainable AI (XAI)** heatmaps. By showing the physician *why* a model chose a diagnosis, we bridge the trust gap.

3. Inherent Error Rates & Clinical Safety

- **Risk:** Deep learning models naturally possess a 5–10% error rate, which can be dangerous in medical triage.
- **Limitation:** This system is a **Decision Support Tool**, not a replacement for a doctor. All high-confidence diagnoses must be signed off by a healthcare professional.

13. References

1. MobileNetV2: Inverted Residuals and Linear Bottlenecks

- *Context:* The core architecture used for high-efficiency medical imaging on local hardware.
- [Link to Paper \(arXiv\)](#)

2. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization

- *Context:* The technical foundation for our XAI heatmaps to resolve the "Visual Context Unavailable" error.
- [Link to Paper \(IEEE\)](#)

3. Multimodal Medical Data Fusion

- *Context:* Research supporting the Late Fusion approach used to combine symptoms and images.
- [Link to Research \(Information Fusion Journal\)](#)

4. Efficient Deep Learning for Healthcare in Resource-Constrained Environments

- *Context:* Addresses the Pakistan 'Triage Gap' and hardware optimization.
- [Link to Paper \(Nature Scientific Reports\)](#)

Student Declaration

We hereby declare that the above proposal is original and has not been copied from any other source.

Student(s) Signature: _____

Supervisor's Signature: _____